Complex Numbers and Trigonometry I

1. Compute

$$(1+i)^{-1},\ (1+2i)^2,\ \frac{3}{1-4i},\ \sqrt{i}.$$

2. Express

$$3,\ -2+2i,\ 1-i\sqrt{2},\ 1+i\sqrt{2}$$

in polar form.

3. Let $z = a + bi$. Show that

$$\operatorname{Re} z = \frac{z+\overline{z}}{2},\ \operatorname{Im} z = \frac{z-\overline{z}}{2i}.$$

Re z denotes the real part of z and Im z the imaginary. For example, if $\gamma = \alpha + \beta i$, then $\operatorname{Re}\gamma = \alpha$, $\operatorname{Im}\gamma = \beta$.

4. For two complex numbers z_1 and z_2, show that

$$\overline{z_1+z_2} = \overline{z_1} + \overline{z_2},\ \overline{z_1 z_2} = \overline{z_1}\cdot\overline{z_2}.$$

5. For two complex numbers z_1 and z_2, find

$$|z_1+z_2|^2,\ |z_1-z_2|^2.$$

6. (a) Show that $\operatorname{Re} z \le |z|$ for any complex number z. **(b)** Using problem 5 and part (a) of this problem, show that

$$|z_1+z_2| \le |z_1| + |z_2|.$$

7. Using the Cartesian distance formula

$$d(p,q) = \sqrt{(p_x-q_x)^2+(p_y-q_y)^2},$$

show that the distance between two points $v_1 = (r_1, \theta_1)$ and $v_2 = (r_2, \theta_2)$ on the polar coordinates is given by

$$d(v_1,v_2) = \sqrt{r_1^2+r_2^2-2r_1r_2\cos(\theta_2-\theta_1)}.$$

This result is also known as the law of cosines.

8. Let $z = x + iy$ and $\gamma = a + bi$. Let $r > 0$ be a real number. Express

$$|z-\gamma| = r$$

only in terms of x, y, a, b, and r. Which geometric figure does this equation represent?

9. Let v be the unit vector $(1, 0)$. Let α and β be two nonzero (or non-$2\pi n$) angles. Rotate v by α and $\alpha + \beta$ to obtain v' and v'', respectively. **(a)** Drop a perpendicular from v' to the positive x-axis. Label the resulting right triangle T_1. What are its side lengths? **(b)** Drop a perpendicular from v'' to the segment from the origin to v'. Label the resulting right triangle T_2. What are its side lengths? **(c)** Finally, drop a perpendicular from v'' to the positive x-axis to form T_3. Using this construction, show that

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \sin\beta\cos\alpha,$$
$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta.$$

10. Using the results from 9, show that for two complex numbers $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$,

$$\arg(z_1 z_2) = \arg z_1 + \arg z_2.$$

What does this result indicate geometrically?